29/8/25

CS 310

① FSA ✓

Closure

Pumping Lemma

② PDA → accept by — empty stack
— final state

DPDA vs NPDA

Minimize

Product Automata

DFA M_1 6 states

DFA M_2 7 states

Given L_1 L_2

regular

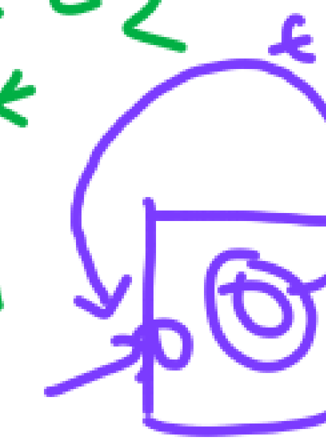
✓ Complement $\overline{L_1}$

DFA + FLIP L_1

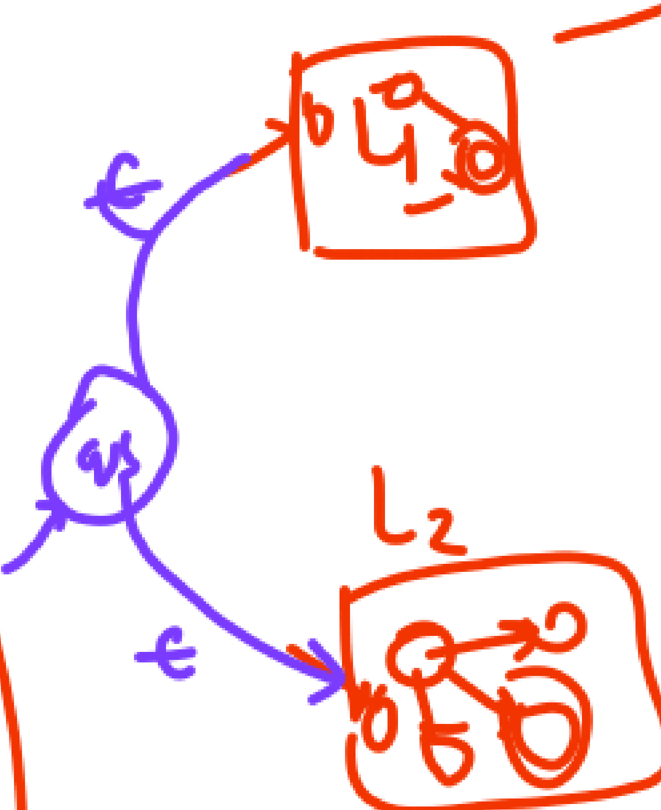
✓ Union $L_1 \cup L_2$

✓ Intersection $L_1 \cap L_2$

** ✓ Kleene Closure L_1^*



✓ Concat. $L_1 ; L_2$



Difference $L_1 - L_2$

Symm. Diff

Reverse L_1^R

{ Half
odd
Even
⋮
Alternate

Pumping Lemma

$L = a^n b^n \mid n \geq 0$ is not regular

Proof-by Contradiction

You | Adversary

A

B

1 → L is regular

~~I then~~

2 → Please build DFA & tell me how many states

348

N-states

3 → Split $a^N b^N$

loop

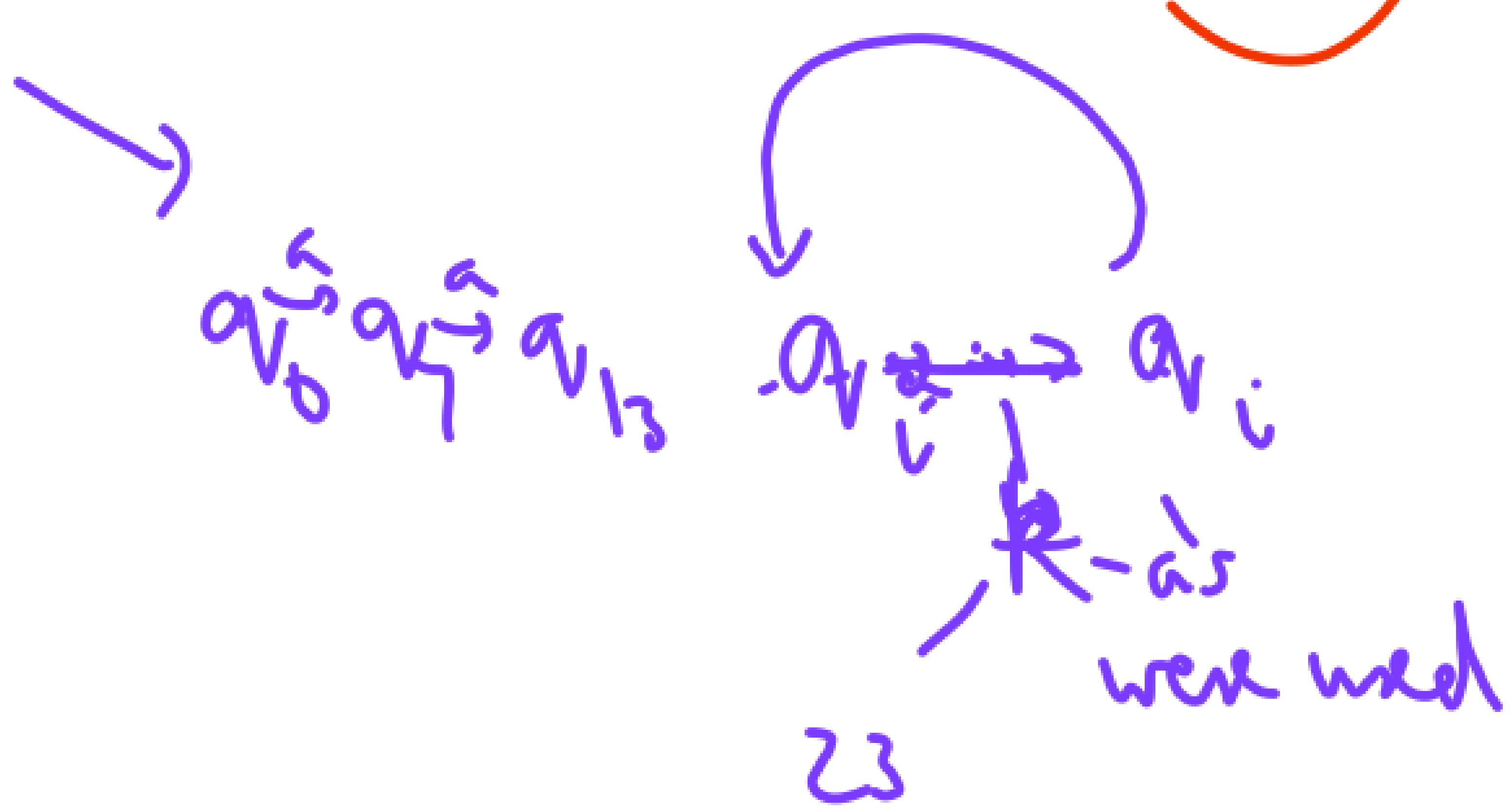
$u v w$

repeats state

$a^{400} b^{400}$

$|uv| \leq N$

Then $u v^i w \in L$ for $i \geq 0$



PDA

$L = \{ a^n b^m a^n \mid n, m \geq 0 \}$

final state

$(q_0, w, Z_0) \vdash \cdots \vdash (q_f, \epsilon, \alpha)$

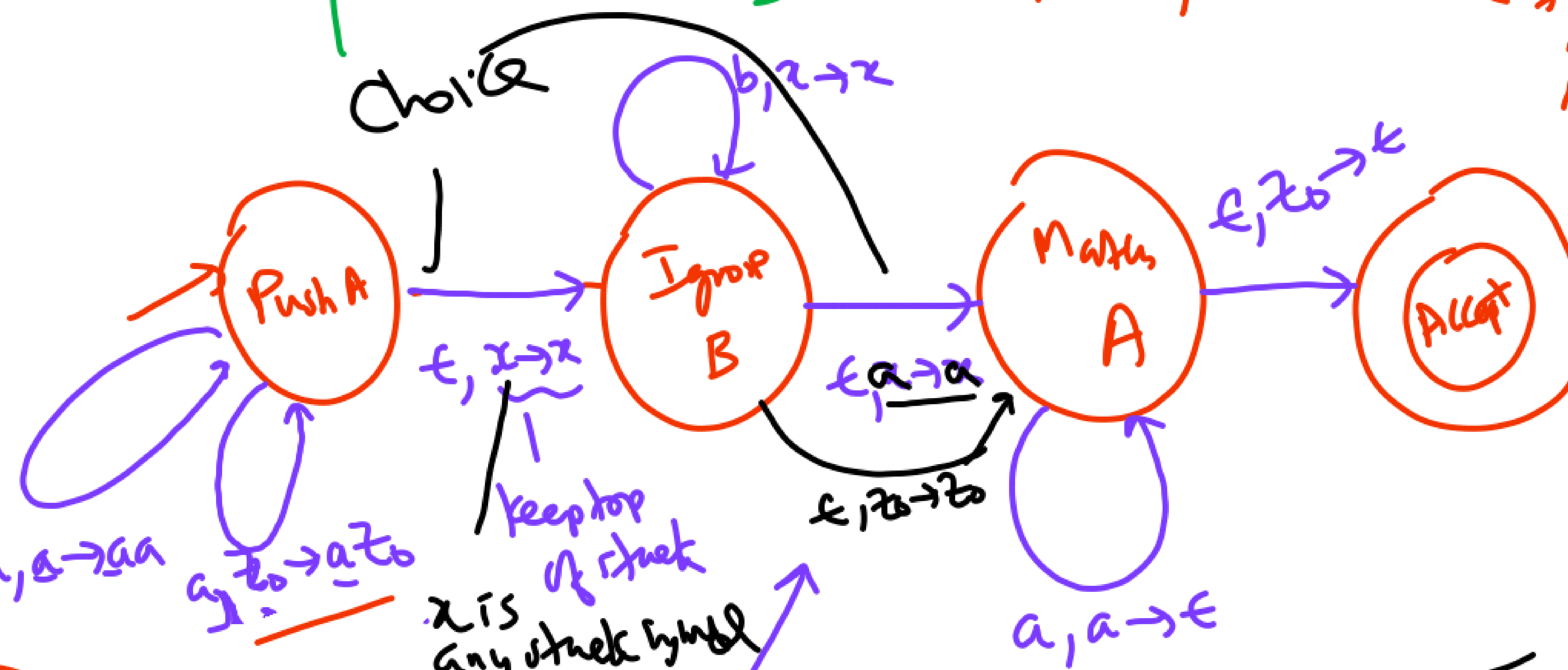
1 $S \to T$ v_1

2 $S \to aSa$ v_2

$T \to bT$

$T \to \epsilon$

v_3



NPDA → Non-deterministic

(6-states)

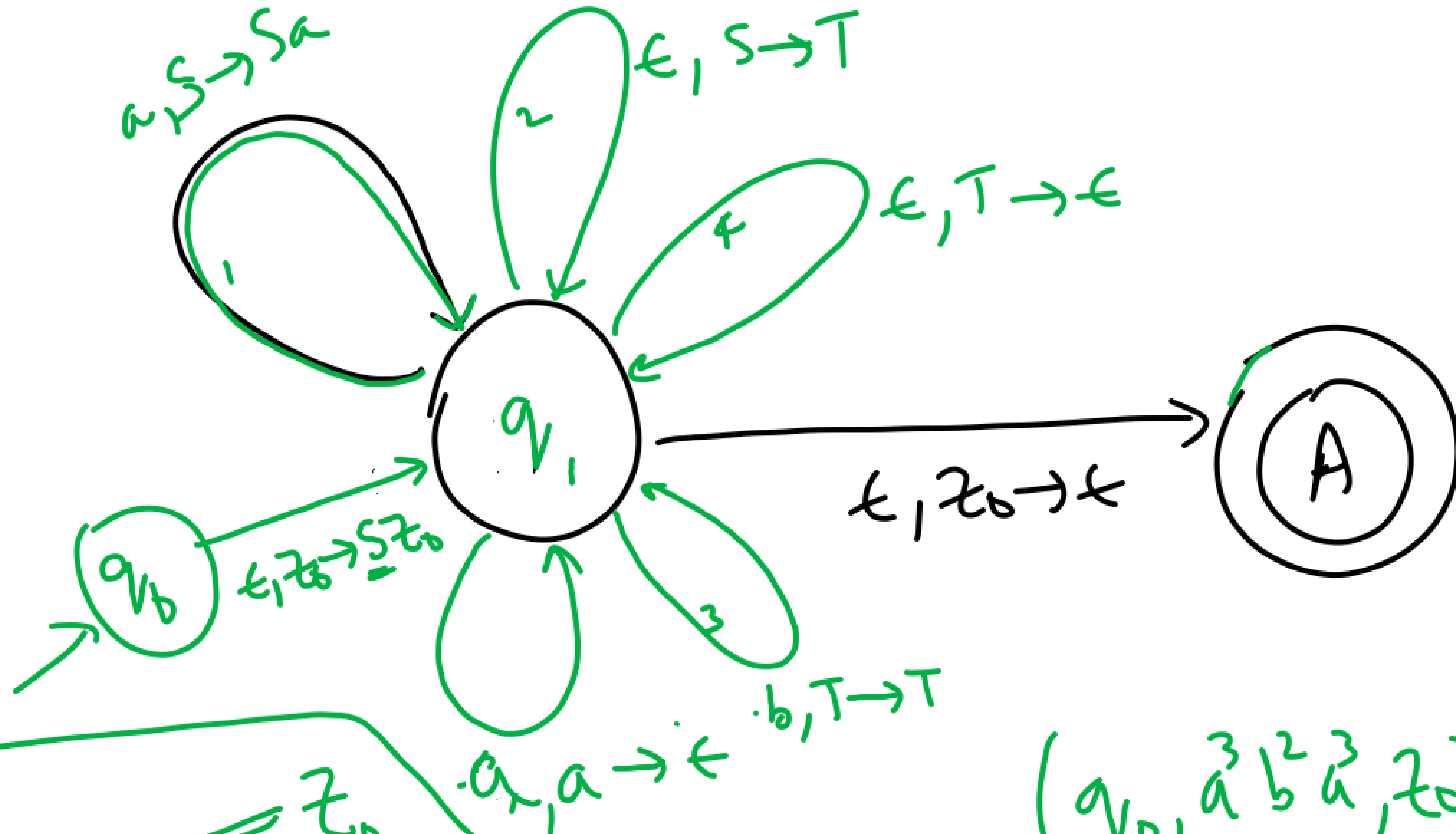
atmost one choice

PDA Deterministic

exactly 1 choice

$(q_0, a^3 b^2 a^3, Z_0) \vdash (q_0, a^2 b^2 a^3, aZ_0)$

$\vdash \quad \vdash \quad \vdash (q_0, b^2 a^3, a^3 Z_0) \vdash (q_1, b a^3, a^3 Z_0)$



Σ

alphabet

z_0

1. $S \to aSa$
2. $S \to T$
3. $T \to bT$
4. $T \to \epsilon$

(q_0, ω, z_0)

bottom of stack

$(q_0, a^3 b^2 a^3, z_0)$

$\vdash$

$\vdash$

$\vdash (A, \epsilon, \epsilon)$